

BATMAN: Blockchain-Based Aircraft Transmission Mobile Ad Hoc Network



Arushi Arora and Sumit Kr Yadav

Abstract Automatic Dependent Surveillance–Broadcast (ADS-B) systems is a recent aviation technology, which has established its standing in the domain of communication between aircrafts and has almost replaced the conventional radar system for transmission of messages. Various loopholes related to privacy and security has been found in ADS-B, by researchers such as message broadcast tampering. In this paper, we propose a methodology based on BATMAN algorithm, which secures the communication between aircrafts and ground stations. This framework for authentication is a novel network topology based on a decentralized blockchain. The authentication is done using a three-way handshake between On-Air Unit (OAU) and Ground Station (GS). The proposed technology involves a group formation of authenticated mobile nodes for communication using the distributed public ledger and further utilizing its security features. The impact of BATMAN algorithm is that it further secures AANET (Aircraft Ad-hoc Network) for ADS-B system.

Keywords ADS-B · Blockchain · ATC · Privacy · Security · Ledger · HashMaps

1 Introduction

The future of transportation system beholds Air Traffic Control (ATC) as its cornerstone. Hartsfield–Jackson Atlanta International Airport has been the world’s busiest airport by passenger traffic since 1998 handling a total of 882,497 aircraft movements in 2015 [24]. This augmentation of the air traffic load is advancing at a tremendous pace. Automatic Dependent Surveillance–Broadcast (ADS-B) is a genre of Air Traffic Management and Control (ATM/ATC) Surveillance system. It is a technology that will soon replace that the conventional radar-based systems [18]. Along with

A. Arora (✉) · S. K. Yadav

Indira Gandhi Delhi Technical University for Women, Kashmere Gate, Delhi, India
e-mail: arushi1250@gmail.com

S. K. Yadav

e-mail: sumit@igdtuw.ac.in

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ADS-B, there exists two broadcast protocols Flight Information Service (FIS-B) and Traffic Information Service (TIS-B), which are used to send the reply back to the On-Air Unit (OAU) from the Ground Station (GS). New threats are emerging along with this technology which forms an integral component of Next Generation Air Transportation System. There is no methodology for entity authentication which adverts one of the serious peril. This does not safeguard immunization from illegitimate messages advancing from third parties or uncertified units making it hackable [4]. Also, these messages can be effortlessly tinkered as they lack the use of data signatures for authentication. One serious repercussion of message encryption dearth is eavesdropping. As a consequence of these momentous privacy and security vulnerabilities, messages can be easily spoofed putting many lives at risk [19]. For example, an aircraft can be forced to reroute its path when a spoofed message informs it of another aircraft flying nearby at the same altitude. Scrutinizing of these messages is possible by conferring the backup radar signal. But this approach manifests to be unfeasible as it consumes plenty of time and energy in a situation wherein a decision has to be made within seconds. Also, there have been some instances of hijacked emergency signals [14]. Federal Aviation Administration (FAA) has also asserted the need of the hour to avert hackers targeting aircraft and air traffic control systems [12]. Another safety alert was issued by Britain's Civil Aviation Authority [21]. Previously, an unknown voice broke into the communication system of US Air flight approaching Washington's Reagan National Airport putting the lives of passengers at risk. The message caused confusion for the pilot of this plane along with the pilots of two aircrafts in the airspace [13].

The blockchain is one of the most ingenious and promising technologies of the future which can overcome the above-stated encumbrances [6]. It is effectively replacing the current transaction system. The idea of blockchain was pioneered by a researcher with a pseudonym "Satoshi Namakato" who applied this technology for implementing the cryptocurrency—Bitcoin [15]. The technology is decentralized and uses a distributed public ledger in which blocks are encrypted and chained together in a chronological order. All the nodes in this framework have a copy of the public ledger and are connected on a peer-to-peer basis [8]. A pair of a public and private key is associated with each node of the network. A block, which is the basic building unit of the chain, stores transactions at a particular timestamp. It encompasses the transactions, its hash value, timestamp, a signature of the block, and nonce [3, 20]. The blockchain is a decentralized which is verifiable and transparent [17]. The records can be altered only if more than 51% of the nodes are in control of the hacker which is unsustainable. In [1, 2, 22, 23] algorithm based on machine learning were used to solve various existing real-world problems. The technology is autonomous and it maintains the anonymity of the sender and receiver in the transaction by utilizing public and private keys of the nodes [16].

In this paper, BATMAN, a scheme to secure communication between aircrafts and ground stations is proposed. Using a public ledger, i.e., a blockchain, authenticated nodes form a group and are allowed for communication. BATMAN prevents any foreign entity which is not a part of that group, to send false messages or to tamper them. The security features of blockchain are further utilized in this mobile network.

The approach is discussed in detail under Sect. 3 of this paper. In Sect. 2 below, we discuss the related work.

2 Related Work

The blockchain is one of the most propitious technologies of the future. In its germinal state itself, the technology has successfully replaced economic transaction system in various organizations. The technology beholds the power to revolutionize the way data is stored or exchanged using an incorruptible public ledger. In addition to its salient features which include immutability, validation, decentralization, and transparency, blockchain promises to provide privacy and security at all points of time. Automatic Dependent Surveillance–Broadcast (ADS-B) systems is a recent aviation technology, which has established its standing in the domain of communication between aircrafts and has almost replaced the conventional radar system for transmission of messages [10]. The ADS-B system uses the standard Global Navigation Satellite System (GNSS) and the traditional communication system between the aircraft and the fixed reference, i.e., Air Traffic Control (ATC). These figures are the broadcasted to the open space. Other ADS-B enabled and the ground ATC are capable to use this information for traffic control in real time. There are various ambiguities that come into consideration in a full-scale implementation of ADS-B. The loopholes that exist are defined in the research paper [25]. This study aims at discovering the drawbacks of implementation of ADS-B and raising awareness regarding the security to oscillate between the technological advancement and the increasing threat magnitude. There are various ambiguities that come into consideration in a full-scale implementation of ADS-B. The paper [5] explains the breakdown of ADS-B protocol in terms of security and attacks on these devices. Ground reality of ADS-B with all its pros and cons is described in [19], but it could not suggest a feasible solution to the problems risen by the ADS-B. In this paper, we propose a synthesis of ADS-B technology and the blockchain protocol so as to overcome all the drawbacks of the ADS-B technique. A blockchain protocol can be defined as a continually expanding of the chain of blocks which are linked and secured with the ciphertext [9, 11]. Each block contains transaction information and is linked to a hash function of the previous block. The blockchain protocol finds its implementation presently in the Bitcoin concept which is a peer-to-peer transaction of digital money. The paper [7] explains how blockchain is implemented.

3 Proposed Work

In this section, we propose a scheme BATMAN, to secure the broadcast of ADS-B messages using blockchain. The concept is explained in detail using three algorithms where all nodes of the group are interconnected on a P2P basis. In the first algorithm,

the aircraft registers with the Air Traffic Control (ATC) for the first time. The aircraft submits its identification details, such as FNO_i Flight no, Airlines ID AID_i , Electronic License No., i.e., ELN_i , Country ID, i.e., CID_i , and other required identification details. The ATC assigns a Pseudo ID (PID_i), Public–Private key pair, namely PK_i and SK_i to the aircraft and stores the mapping of the actual identity with the assigned PID_i , and the corresponding public key for that PID_i .

This entry forms a transaction of the $Block_i$ digitally signed by the ATC and added to the identity ledger β with other necessary descriptors. The hash is computed and result forms the $HPrev$ for the next block. $Block_i$ is uploaded to ledger β . The issued PID_i , a copy of the identification ledger entry in β and a pointer PTR_i to the $Block_i$ along with the Private Key SK_i for the issued PID_i are stored with GS. The structure of public ledger β is shown in Fig. 1. Table 1 lists the keys used in the algorithms.

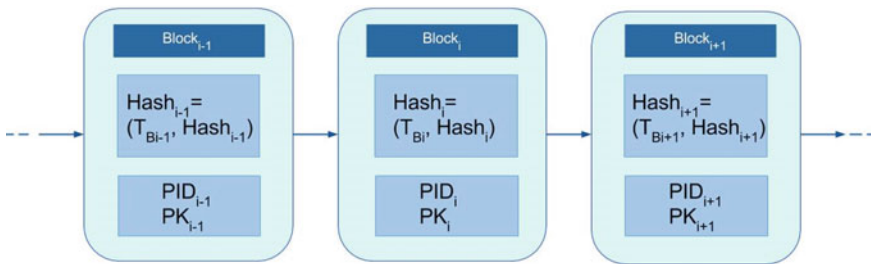


Fig. 1 The structure of public ledger β

Table 1 Keys used in the algorithm

Key	Description	Key	Description
OAU_i	ith On-Aircraft Unit	PID_i	Pseudo ID assigned by ATC
GS	Ground Station	DS_{ATC}	Digital Signature of ATC
FNO_i	Flight Number of OAU_i	$H()$	Hashing function
AID_i	Airlines ID of OAU_i	$Block_i$	Block of the public ledger
G_k	Group public key	$HPrev_i$	Hash of the previous block
CER_{GS}	Certificate given to GS by the ATC	T_{Bi}	Transaction added to the ith block
ID_i	Original Identity given to the ATC	PTR_i	Pointer to the ledger copy
G_m	Group of nodes	CID_i	Country ID of OAU_i
G_n	Group of nodes	ID_{GS}	ID of Ground Station
SK_i	Private key of ith OAU	PK_i	Public key of ith OAU
S_{GS}	Private Key of the GS	DS_{GS}	Digital Signature of GS
B	Public Ledger for authentication; resides with all GSs	ELN_i	Electronic License Number of OAU_i
HM	HashMap	M_i	Message
K_{AOU_i}	Key stored in AOU_i	GS_k	kth Ground Station

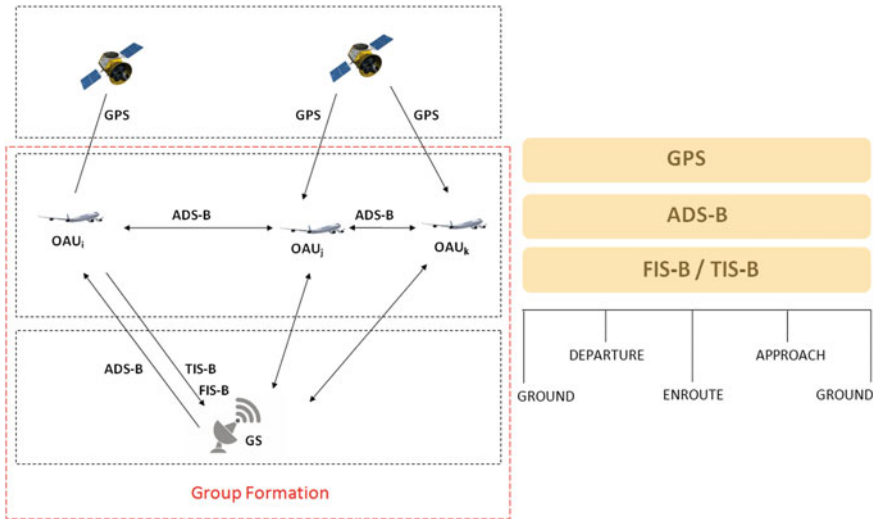


Fig. 2 Group formation after authentication of AOU's

The second algorithm explains the authentication of OAU with GS when they encounter for the first time. When the OAU_i is on the ground (before the flight), it authenticates itself with the GS with a three-way handshake and becomes part of the group of aircrafts in range of the GS. For this, some initial formalities are performed. The OAU_i receives a message containing the GS's certificate CER_{GS} from an in-range GS, it sends a three-tuple message M_i containing its PID_i , ledger copy, and pointer details encrypted with ID_{GS} . The GS upon receiving the message decrypts it using its private key S_{GS} , and gets the PID_i , corresponding to the ledger entry and pointer to the block. It searches for the block decrypts the transaction entry with PK_{ATC} and obtains the corresponding public key PK_i . To verify the identity of the OAU_i, as it is who it claims to be, it encrypts a challenge using the PK_i which can only be decrypted if it has the corresponding private key SK_i . When the OAU_i is able to decrypt using the SK_i , now it sends a response to the challenge after which it completes the mutual authentication step. Upon confirming the identity and authentication of the AOU_i, it now stores the PID_i and other necessary details, in a HashMap to convey to other GSs for authenticating this PID to avoid the above steps for authentication. It simultaneously sends the copy of mapping entry digitally signed with its private key S_{GS} which gets stored in AOU_i. Also, GS sends the group public key G_k , which the GS uses to broadcast messages to G_n . The group formation is shown in Fig. 2.

The OAU will encounter various GS right from its departure to landing. In order to tackle the situation of repeated authentication with simultaneously changing GSs after a period of time, we define the third algorithm. When the aircraft moves from the coverage of one GS to another GS, it need not perform the above handshake as it has already been authenticated by one GS. As the aircraft AOU_i enters the coverage

Algorithm 1:	Registration of the aircraft (AOU_i)
Input: Aircraft's identification information, public ledger β	
1.1. $\text{AOU}_i \rightarrow \text{ATC}\{\text{Details: FNo}_i, \text{AID}_i, \text{ELN}_i, \text{CID}_i\}$	
1.2. $\text{MAP}(\text{ID}_{i(\text{ATC})}, \{(\text{PID}_i, \text{PK}_i) \text{DS}_{\text{ATC}} \})$	
1.3. $\text{T}_{\text{Bi}}: \{(\text{PID}_i, \text{PK}_i, \text{other descriptors}) \text{DS}_{\text{ATC}} \}$	
1.4. $\text{H}(\text{T}_{\text{Bi}}, \text{HPrev})$	
1.5. Upload(T_i, β)	
1.6. $\text{GS} \leftarrow \{ \text{PID}_i, \text{SK}_i, \text{PTR}_i, \beta \}$	
Output: Addition of an entry corresponding to a new aircraft in authentication ledger β	
Algorithm 2:	Authentication: OAU's first encounter with a GS
Input: Aircraft's identification information, public ledger β	
2.1. $\text{GS} \rightarrow \text{OAU}: \{ \text{CER}_{\text{GSi}} \}$	
2.2. $\text{M}_i = \{ (\text{PID}_i, \beta, \text{ID}_{\text{GS}}, \text{PTR}_i) \}$	
2.3. Encrypt ($\text{ID}_{\text{GS}}, \text{M}_i$)	
2.4. $\text{OAU} \rightarrow \text{GS}: \{ \text{M}_i \}$	
2.5. $\text{GS} \rightarrow \text{Decrypt}(\text{S}_{\text{GS}})$	
2.6. $\text{GS} \rightarrow \text{Search}(\beta, \text{PK}_{\text{ATC}})$	
2.7. $\text{GS} \rightarrow \text{Encrypt}(\text{PK}_i, \text{Challenge})$	
2.8. $\text{OAU} \rightarrow \text{Decrypt}(\text{SK}_i, \text{Challenge})$	
2.9. $\text{K}_{\text{AOUi}} = \text{DS}_{\text{GS}}(\text{S}_{\text{G}})$	
2.10. $\text{HM} = (\text{PID}_i, \text{K}_{\text{AOUi}})$	
2.11. $\text{AOU}_i \rightarrow \text{Store}(\text{K}_{\text{AOUi}})$	
2.12. $\text{GS} \rightarrow \text{Store}(\text{PID}_i)$	
2.13. $\text{GS} \rightarrow \text{AOU}_i(\text{G}_k)$	
2.14. $\text{G}_n = \text{G}_n \cup (\text{OAU}_i)$	
Output: First time authentication AOU _i and membership with Ground Station GS	
Algorithm 3:	Authentication with changing GS (GS_k)
Input: Aircraft's identification information, public ledger β	
3.1. $\text{OAU}_i \rightarrow \{ (\text{PID}_i, \text{ID}_{\text{GSi}}, \text{CER}_{\text{GSi}}) \text{DS}_{\text{GSi}} \}$	
3.2. $\text{G}_m = \text{G}_m \cup (\text{OAU}_i)$	
3.3. $\text{GS}_k \rightarrow \text{AOU}_i(\text{G}_k)$	
Output: Authentication AOU _i and membership with changing Ground Station GS	

of another GS, it sends the authentication copy by previous GS, which is digitally signed by the previous GS. Since all GSs are assumed to form a network, thus upon receiving a network joining request from PID_i , it just verifies with previous GS, which searches the HashMap and returns a true value for this PID_i , which is then made part of the group in-range G_m of new GS_k and given new Group key G_k .

4 Conclusion and Future Work

As there has been always debate and news related to the security of communication link in the discipline of aviation, it becomes evident that there is a need for more secure forms of communication. ADS-B is the latest technique developed to overcome the conventional radar communication, though it has its own share of disadvantages and

loose ends. Blockchain protocol was initially developed to ensure security in the P2P transfer of Bitcoins, but its startling performance has opened doors for implementation in relatively less secure areas. In this paper, we proposed a scheme BATMAN, to secure the broadcast of ADS-B messages using blockchain technology. Formation of a group by allowing authenticated nodes to enter secures the network from various malicious activities like tampering with messages or sending false signals. Various security issues related to ADS-B can be successfully overcome using BATMAN. From the above-proposed algorithm, we can conclude that an approval and implementation of ADS-B technique along with the infusion of blockchain concept for security management can lead to a significant measure of security and safety. Blockchain indeed is the technology of the future. A lot of financial multinationals and banks are working to handle transactions using blockchains. The blockchain technology is on its way reinventing the way we work and live. It eliminates the use of centralized devices in IoT (Internet of Things), cybersecurity, and networking. The technology provides a new way of managing trust and can be effectively applied in insurance and domains like finance. It eliminates the involvement of third party, hence finding its effective utilization in private transport and ride sharing.

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